Faculty of Science
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ConstructPro ERP

Test Case Register

Requirements traced to automated tests

SENG 34213 – System Development Project

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Deliverable for documents/testing/test-register.pdf (guideline §10.1)

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Chapter 1

Overview

This register lists the test cases that trace the SRS requirements to the automated test suites of ConstructPro ERP, in the format of Section 6.3.3 of the SENG 34213 guideline. It is the same register as Appendix A of the Final Development Report.

Results marked **Pass** come from the CI run on `develop` used in the report (backend `f6bf853`: 1,535 unit tests and 548 integration tests). End-to-end cases marked **Not run** have not yet been executed against the deployed system, because the deployed Playwright workflow is started by manual dispatch and has not been run. Of the 23 cases, 13 pass and 10 are not yet run.

Table 1.1: Test cases by type and status

Type	Cases	Pass	Not run
Unit	8	8	0
E2E	10	0	10
Integration	5	5	0
Total	23	13	10

Table 1.2: Test case summary

Test Case ID	Test Case Name	Type	Status
TC-FR-001-01	Login issues access and refresh tokens	Unit	Pass
TC-FR-001-02	Create user, assign role, no password in response	E2E	Not run
TC-FR-001-03	Reject duplicate user email	Unit	Pass
TC-FR-002-01	Capture, search, edit and delete a lead	E2E	Not run
TC-FR-003-01	Quotation total calculated on the server	E2E (API)	Not run

Test Case ID	Test Case Name	Type	Status
TC-FR-003-02	Live total updates while adding items	E2E (UI)	Not run
TC-FR-003-03	Rejection requires a reason	E2E (UI)	Not run
TC-FR-004-01	Duplicate conversion is blocked	E2E (API)	Not run
TC-FR-004-02	Concurrent conversions create one project	Integration	Pass
TC-FR-005-01	Weighted progress uses relative weights	Unit	Pass
TC-FR-005-02	Invalid milestone weight rejected	Unit	Pass
TC-FR-005-03	Project cannot complete with incomplete milestones	Unit	Pass
TC-FR-006-01	Duplicate and excessive payments rolled back	Integration	Pass
TC-FR-006-02	Payment updates invoice and dashboard KPIs	E2E	Not run
TC-FR-007-01	Revenue KPIs from invoice aggregates	Unit	Pass
TC-FR-008-01	Oversized document upload rejected	Integration	Pass
TC-FR-008-02	Unauthenticated document request rejected	Integration	Pass
TC-FR-009-01	Calendar refresh token stored encrypted	Unit	Pass
TC-FR-010-01	AI prediction falls back safely	Unit	Pass
TC-FR-010-02	AI prediction shown in Analytics	E2E	Not run
TC-NFR-08-01	Client portal hides other customers' data	Integration	Pass
TC-NFR-08-02	Non-finance role cannot use finance APIs	E2E (Security)	Not run
TC-NFR-08-03	Accountant cannot list users or leads	E2E (Security)	Not run

Before submission

After the deployed Playwright run, fill in Actual Output and Status for every “Not run” case, and record the date of the run and the passed, failed and skipped counts here.

Chapter 2

Test Cases

Each test case uses the fields of Section 6.3.3 of the guideline.

Field	Content
Test Case ID	TC-FR-001-01
Test Case Name	Login issues access and refresh tokens
Related Requirement	FR-001 (Manage Users), NFR-07 (JWT Authentication)
Related Test File	`backend/test/auth-service/google-session.spec.ts`
Type	Unit
Priority	P1 – Must Pass
Preconditions	Active user exists
Input	Valid sign-in
Expected Output	Access token valid 15 min and a separate refresh token valid 7 days
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-001-02
Test Case Name	Create user, assign role, no password in response
Related Requirement	FR-001 (Manage Users), UC-01
Related Test File	`tests/e2e/user-management.spec.ts`
Type	E2E
Priority	P1 – Must Pass
Preconditions	Logged in as ADMIN
Input	New user form with role
Expected Output	User created with role; response has no password field
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-001-03
Test Case Name	Reject duplicate user email
Related Requirement	FR-001 (Manage Users)
Related Test File	`backend/test/user-service/user.service.spec.ts`
Type	Unit
Priority	P2 – Should Pass
Preconditions	User with the email exists
Input	Create user with same email
Expected Output	409 conflict; original account unchanged
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-002-01
Test Case Name	Capture, search, edit and delete a lead
Related Requirement	FR-002 (Manage Sales Leads), UC-02
Related Test File	`tests/e2e/leads.spec.ts`
Type	E2E
Priority	P1 – Must Pass
Preconditions	Logged in as SALES_MANAGER or ADMIN
Input	Lead details, search term, note
Expected Output	Lead persists through each step and is removed on delete
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-003-01
Test Case Name	Quotation total calculated on the server
Related Requirement	FR-003 (Generate Quotations), UC-03
Related Test File	`tests/e2e/quotations/quotations-backend.spec.ts` (BE-01)
Type	E2E (API)
Priority	P1 – Must Pass
Preconditions	Lead exists
Input	POST /quotations with three line items
Expected Output	201; total equals the sum of quantity × unit price
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-003-02
Test Case Name	Live total updates while adding items
Related Requirement	FR-003 (Generate Quotations)
Related Test File	`tests/e2e/quotations/quotations-frontend.spec.ts` (FE-02)
Type	E2E (UI)
Priority	P2 – Should Pass
Preconditions	Quotation form open
Input	Add and edit line items
Expected Output	Displayed total updates immediately
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-003-03
Test Case Name	Rejection requires a reason
Related Requirement	FR-003 (Generate Quotations), BR 10.2
Related Test File	`tests/e2e/quotations/quotations-frontend.spec.ts` (FE-06)
Type	E2E (UI)
Priority	P2 – Should Pass
Preconditions	Quotation PENDING_APPROVAL
Input	Reject without a reason
Expected Output	Validation error; status unchanged
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-004-01
Test Case Name	Duplicate conversion is blocked
Related Requirement	FR-004 (Convert Quotation to Project), BR 10.2
Related Test File	`tests/e2e/quotations/quotations-backend.spec.ts` (BE-07)
Type	E2E (API)
Priority	P1 – Must Pass
Preconditions	Quotation already CONVERTED
Input	Approve and convert again
Expected Output	409 ALREADY_CONVERTED with the existing project ID
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-004-02
Test Case Name	Concurrent conversions create one project
Related Requirement	FR-004 (Convert Quotation to Project), UC-04
Related Test File	`backend/test/project-service/project-service.integration.spec.ts`
Type	Integration
Priority	P1 – Must Pass
Preconditions	Approved quotation
Input	Two simultaneous conversion requests
Expected Output	Both return the same project; one project row exists
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-005-01
Test Case Name	Weighted progress uses relative weights
Related Requirement	FR-005 (Track Project Milestones), UC-05
Related Test File	`backend/test/project-service/milestone.service.spec.ts`
Type	Unit
Priority	P1 – Must Pass
Preconditions	Project with milestones of different weights
Input	Update milestone progress
Expected Output	Project progress = Σ(progress × weight) / Σ weight
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-005-02
Test Case Name	Invalid milestone weight rejected
Related Requirement	FR-005 (Track Project Milestones)
Related Test File	`backend/test/project-service/milestone.controller.spec.ts`
Type	Unit
Priority	P2 – Should Pass
Preconditions	Project exists
Input	Weight 0 or 11
Expected Output	400 validation error
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-005-03
Test Case Name	Project cannot complete with incomplete milestones
Related Requirement	FR-005 (Track Project Milestones), BR 10.4
Related Test File	`backend/test/project-service/project.service.spec.ts`
Type	Unit
Priority	P1 – Must Pass
Preconditions	ACTIVE project, one milestone IN_PROGRESS
Input	Set status COMPLETED
Expected Output	Rejected with a lifecycle error
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-006-01
Test Case Name	Duplicate and excessive payments rolled back
Related Requirement	FR-006 (Track Payments), BR 10.3
Related Test File	`backend/test/finance/finance-workflow.db.spec.ts`
Type	Integration
Priority	P1 – Must Pass
Preconditions	ISSUED invoice with outstanding balance
Input	Payment above balance; repeated reference
Expected Output	422 / 409; persisted balances unchanged
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-006-02
Test Case Name	Payment updates invoice and dashboard KPIs
Related Requirement	FR-006 (Track Payments), FR-007 (KPI Dashboard), UC-06
Related Test File	`tests/e2e/finance-dashboard.spec.ts`
Type	E2E
Priority	P1 – Must Pass
Preconditions	Logged in as ACCOUNTANT; ISSUED invoice
Input	Record partial payment
Expected Output	Invoice PARTIALLY_PAID, balance reduced, KPIs and UI consistent
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-FR-007-01
Test Case Name	Revenue KPIs from invoice aggregates
Related Requirement	FR-007 (KPI Dashboard), UC-07
Related Test File	`backend/test/analytics-service/analytics.service.spec.ts`
Type	Unit
Priority	P2 – Should Pass
Preconditions	Invoices and payments exist
Input	GET revenue KPIs
Expected Output	Totals match invoice aggregates
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-008-01
Test Case Name	Oversized document upload rejected
Related Requirement	FR-008 (Store Project Documents), UC-08
Related Test File	`backend/test/document-service/document-gateway.e2e-spec.ts`
Type	Integration
Priority	P2 – Should Pass
Preconditions	Logged in as PROJECT_MANAGER
Input	File larger than the limit
Expected Output	413 with a document-specific error code
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-008-02
Test Case Name	Unauthenticated document request rejected
Related Requirement	FR-008 (Store Project Documents), NFR-08 (Role-Based Access Control)
Related Test File	`backend/test/document-service/document-gateway.e2e-spec.ts`
Type	Integration
Priority	P1 – Must Pass
Preconditions	None
Input	Request without token
Expected Output	401 TOKEN_MISSING
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-009-01
Test Case Name	Calendar refresh token stored encrypted
Related Requirement	FR-009 (Google Calendar Integration), UC-09
Related Test File	`backend/test/calender-service/calendar.service.spec.ts`
Type	Unit
Priority	P1 – Must Pass
Preconditions	User completes Google consent
Input	Store connection
Expected Output	Token stored as AES-256-GCM ciphertext with IV and tag
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-010-01
Test Case Name	AI prediction falls back safely
Related Requirement	FR-010 (AI Risk Prediction), UC-10
Related Test File	`backend/test/ai-service/ai-forecasting.service.spec.ts`
Type	Unit
Priority	P2 – Should Pass
Preconditions	Project with little history
Input	Request risk prediction
Expected Output	SAFE_FALLBACK result with explanation and low confidence
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-FR-010-02
Test Case Name	AI prediction shown in Analytics
Related Requirement	FR-010 (AI Risk Prediction), UC-10
Related Test File	`tests/e2e/ai-analytics.spec.ts`
Type	E2E
Priority	P2 – Should Pass
Preconditions	Logged in as management user
Input	Generate prediction
Expected Output	Contract-valid prediction displayed
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-NFR-08-01
Test Case Name	Client portal hides other customers' data
Related Requirement	NFR-08 (Role-Based Access Control), ADR-15
Related Test File	`backend/test/client-portal/client-portal-ownership.integration.spec.ts`
Type	Integration
Priority	P1 – Must Pass
Preconditions	Two customers with projects
Input	Portal user lists projects and documents
Expected Output	Only own projects returned; mixed-customer projects hidden
Actual Output	As expected
Status	Pass

Field	Content
Test Case ID	TC-NFR-08-02
Test Case Name	Non-finance role cannot use finance APIs
Related Requirement	NFR-08 (Role-Based Access Control), BR 10.1
Related Test File	`tests/e2e/authorization.spec.ts`
Type	E2E (Security)
Priority	P1 – Must Pass
Preconditions	Logged in as SALES_MANAGER
Input	Call invoice and payment endpoints
Expected Output	403 FORBIDDEN
Actual Output	……
Status	Not run

Field	Content
Test Case ID	TC-NFR-08-03
Test Case Name	Accountant cannot list users or leads
Related Requirement	NFR-08 (Role-Based Access Control), BR 10.1
Related Test File	`tests/e2e/management-authorization.spec.ts`
Type	E2E (Security)
Priority	P1 – Must Pass
Preconditions	Logged in as ACCOUNTANT
Input	GET /users, GET /leads, POST /clients
Expected Output	403 FORBIDDEN for each
Actual Output	……
Status	Not run